Asad Ikram

MSc Data Analytics (Distinction), University of Portsmouth · Chevening Scholar 2024/25
Applicant, Graduate Program in Complexity Science, CSH Digital Innovation School (PhD 2027)
Lahore, Pakistan · +44 7482 376417 · asad.ikram53@gmail.com · linkedin.com/in/asad-ikram98 · github.com/asadfix

PROFILE

I study whether machine labels of online text mean the same thing across platforms, models and time. My MSc dissertation analysed public reaction to UK economic policy on four platforms, and a methods paper with Dr Ella Haig, submitted to arXiv in October 2026, brings psychometric tools to machine-labelled sentiment. Since 2018 I have built the data collection and LLM labelling pipelines that analysis of online conversation depends on, and I check machine labels against human ones before anything is built on them.

EDUCATION

Sep 2024 - Sep 2025
awarded 8 Dec 2025

MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Chevening Scholar. Credit-weighted average 70.5/100; master's project 78/100. Supervisor: Dr Ella Haig. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80). Taught and assessed in English.

Dissertation: *Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies.* 255,446 public comments from X/Twitter, Reddit, YouTube and Quora, collected through official APIs and licensed feeds under university ethics approval (TETHIC-2025-111094). Fine-tuned BERT against BiLSTM and lexicon baselines (macro-F1 0.878, scored against lexicon-seeded labels); 200 posts hand-labelled by me, $\kappa \approx 0.78$ against VADER; calibration, McNemar and bootstrap tests, Prophet forecasting, LIME and integrated gradients.

Aug 2017 - Jan 2022

BS Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan; Karachi campus, then Chiniot-Faisalabad campus.
130 credit hours, CGPA 2.75/4.00. Courses included Probability and Statistics, Linear Algebra, Calculus I and II, Numerical Methods, Design and Analysis of Algorithms and Applied Machine Learning.

RESEARCH

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Submitted to arXiv, October 2026 (cs.LG).** Treats each platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, pooling platform scores at face value gives the wrong sign of daily aggregate sentiment on roughly a quarter of days, from measurement differences alone. *No peer-reviewed publications to date.*

EXPERIENCE

Dec 2024 - present
part-time

Data Engineer, M+C Saatchi Fluency, London (remote). Social media analysis for global brands (consumer goods, automotive, sport, e-commerce, health) and UK government.

- Social media data end to end: collecting, labelling and tagging, text and sentiment analysis with AI and ML, forum sourcing, and the system architecture behind the dashboards clients use to decide. This is where I learned the social media industry.
- Analyst on a study of twelve months of social discourse (100,845 Brandwatch and 170,542 Reddit mentions): BERTopic-derived codebook, LLM tagging with a prompt-version hash on every tag, three pilot runs on the same 500 rows before the full run; contributor to LLM-assisted coding of about 170,000 multilingual posts and reviews.
- Data infrastructure: Sprinklr and Brandwatch feeds plus custom crawlers into AWS with SageMaker NLP; over 60 crawlers across ten platforms daily (6.7 million items in six weeks), kept running by a self-healing LLM agent and a builder agent that creates new crawlers; a weekly 2.2-million-row social-analytics pipeline with an 89-check quality sweep.

2025 - present part-time

Co-Founder and Chief Technology Officer, Artemis AI (formerly ViralData), United Kingdom (remote). Social-media intelligence platform for creators and businesses; I lead the technology in a team of eight; beta on 15 December 2026.

- Built the data platform on AWS: Meta Graph API collectors, a layered warehouse queried with Athena, and weekly labelling runs; 111,000+ posts and 3.2 million comments processed so far, with image and video understanding (OCR, CLIP, Whisper).
- Designed an LLM labelling chain: Claude labels about 6,000 posts with an explanation of each, rules mined from the explanations become a prompt for DeepSeek, which labels about 12,000 more to train encoder classifiers (XLM-R, DistilBERT, DeBERTa-v3, Detoxify). The team first hand-labelled about 500 posts and comments to check both models.
- Designing model evaluation around a human-labelled golden set, so every model is checked against people before it reaches users.
- Leading the build for the beta, 48 models in the agreed architecture: Art-E, an advisor on Amazon Bedrock that gives cited answers from each client's own data behind guardrails; crisis alerts with storm detection and spam, bot and banter filters; Instagram ingestion; and models for best posting times, engagement outlook, comment intent and audience communities.

May 2023 - present contract, part-time

Data Engineer (contract), Fix.com, Canada (remote). Online replacement-parts retailer. Since June 2026 designed and built, alone, the company's crawler platform (about 10 million rows across 82 datasets by October 2026).

Sep 2024 - present part-time, remote

Shorter engagements. CXG, Dubai: production crawlers and a pytest QA framework. **VendueTech**, Ireland, Lead Data Engineer, Jun - Jul 2025: led three engineers on an Azure-to-AWS migration. **Prefe**, Milan, Data Engineer, Sep - Dec 2024: moved legacy collectors to Scrapy, cutting the failure rate by 60%; mentored two junior engineers.

Feb 2022 - Aug 2024 full-time

Associate Software Engineer, then Data Engineer, then Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Lahore. Promoted twice in about two years.

- Built competitor-analysis dashboards from e-commerce and real-estate listings and online discussions that predicted revenue and showed what customers said.
- Led a three-engineer team running 500+ scrapers across 15 markets on Kubernetes/EKS, collecting text in English, Arabic, Urdu and Hindi; redesigned the layered warehouse, raising ETL throughput by 60%.

Dec 2018 - Feb 2026 part-time

Freelance data engineer, Fiverr. 300+ completed orders in web scraping, automation, ETL and data analysis; Level Two seller (first reached August 2021).

SKILLS

Methods

Validating machine and LLM labels against hand-labelled sets; macro-F1, Cohen's κ, calibration; McNemar and bootstrap tests; generalizability theory, measurement invariance and DIF, score linking; simulation with known ground truth.

NLP and ML

Fine-tuning encoder classifiers (BERT, XLM-R, DistilBERT, DeBERTa-v3); LLM-assisted labelling through the Claude and DeepSeek APIs; BERTopic; LIME and integrated gradients. I have not fine-tuned a generative model.

Programming and data

Python (Scrapy, PyTorch, pandas, FastAPI), SQL, PySpark; R (functional); Git and CI; AWS (SageMaker, Fargate, Lambda, Athena, Redshift), Docker, Kubernetes; official platform APIs (Meta Graph, Reddit, YouTube).

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening

Chevening Scholar 2024/25, the UK Government's global scholarship, which funded the MSc; awarded for helping students into remote careers and for a cross-platform social media analytics proposal.

Public work

Author and maintainer of web-scraping-guide.com (since May 2026), a free public guide to web data collection, including the legal and ethical limits of collection.

Languages

Urdu (native); English (working language; MSc taught and assessed in English, IELTS 6.5).

REFEREES

Dr Ella Haig, Associate Professor in Artificial Intelligence, School of Computing, University of Portsmouth. ella.haig@port.ac.uk. MSc supervisor and co-author.

Dr Fahad Ahmad, Senior Lecturer, Centre for Cybercrime and Economic Crime, University of Portsmouth. Fahad.Ahmad@port.ac.uk. Taught and assessed me on the MSc.

Portfolio of work

Asad Ikram · A selection of the work I care most about, with why each piece is here. Client work has no public link and is described without client data.

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1. "Is the Platform Part of the Measurement?" Methods paper with Dr Ella Haig

What: a six-step protocol (generalizability theory, invariance and DIF, score linking) for comparing machine-labelled sentiment across platforms, with a 500-replication simulation that reproduces in seconds and needs no collected data. **Why included:** it is the clearest statement of the question I want to pursue, and I built the simulation behind its main result.

Submitted to arXiv, October 2026 (cs.LG; cross-listed to cs.CY, cs.CL, cs.SI). PDF available on request.

2. MSc dissertation: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies (Portsmouth, 2025)

What: 255,446 comments from four platforms under ethics approval, BERT against BiLSTM and lexicon baselines, calibration and significance tests. **Why included:** my first research project end to end, and the one I learned most from, because its validation turned out to be circular (see answer 1). The paper above grew out of that mistake.

Available on request.

3. The LLM labelling chain behind a social-media intelligence platform (Artemis AI)

What: Claude labels about 6,000 posts with an explanation of each; rules mined from those explanations become a prompt for DeepSeek, which labels about 12,000 more to train encoder classifiers, with about 500 hand-labelled posts as the check on both. **Why included:** it is my research question inside a product heading to beta in December: every one of its 48 models rests on these labels, so they have to be checked against people first.

Internal work; product at artemisai.co.uk (pre-launch).

4. Twelve months of social discourse, coded with LLMs and checked by hand (M+C Saatchi Fluency)

What: about 271,000 Brandwatch and Reddit mentions; a codebook derived from BERTopic, LLM tags each carrying a hash of the prompt version, and three pilot runs on the same 500 rows before the full run. **Why included:** it is the closest I have come to studying narratives at scale, and it taught me to treat a prompt as an instrument whose version must be recorded.

Client work; no public link.

5. Self-healing and builder agents for a crawler fleet (M+C Saatchi Fluency)

What: over 60 crawlers of owner forums and social platforms; when a source changes, an LLM agent diagnoses the break, regenerates the spider and tests it on live pages, a second agent builds new crawlers, and a person approves each merge. **Why included:** my favourite piece of engineering. A collector that breaks silently does not stop a time series, it thins it, and that is a measurement problem as much as an engineering one.

Client work; no public link.

6. Multilingual competitor dashboards at Dubizzle Labs (2022 - 2024)

What: dashboards built from listings and online discussion in English, Arabic, Urdu and Hindi across 15 markets that predicted revenue and showed what customers said. **Why included:** my first experience of turning large volumes of public, multilingual text into claims about markets that decision-makers acted on.

Internal work; no public link. Profile: linkedin.com/in/asad-ikram98